

Characterization of Stars in the Habitable Worlds Observatory Target List

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Abstract

The Habitable Worlds Observatory (HWO) is a planned space telescope tentatively set to launch in the 2040s. It will be designed to image and characterize around 25 exoplanets to determine which ones have Earth-like conditions and also search for biosignatures. The HWO project has released a list of 164 potential target stars to study, so it is important to find out as much as we can about these star systems prior to the launch of the HWO to see which are the most worthwhile so we can make the most out of the time on the telescope. In this study, we use astronomy tools such as SIMBAD and Xmatch to characterize the HWO list of stars and see what the similarities and differences are among the individual stars and among the different classifications of stars. We also evaluate the effectiveness of using gyrochronology on the HWO stars. We found that the stars tended to have parameters similar to that of the Sun, with a few variations, and that various graphs of stellar properties tended to match closely with the predicted values, with a few discrepancies. We also found that many of the stars in the sample aren't suitable for using gyrochronology, as there was a mismatch between the calculated and recorded ages, perhaps due to the quality of rotation period data, binarity, other sample-quality issues, or the limitations of the model. Using the information from this study, astronomers working on the HWO project might be able to see what gaps exist in current data, which would be relevant to future characterization.

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Every ten years, astronomers from around the United States come together to plan out the priorities for astronomy research for the next decade. A decadal survey titled “*Pathways to Discovery in Astronomy and Astrophysics for the 2020s*” from the decadal meeting recommended launching a large telescope to search for habitable exoplanets (Astro2020). Thus, the Exoplanet Exploration Program was started at NASA, with the Habitable Worlds Observatory set to launch in 2030. Due to later setbacks, the date was pushed back to the 2040s. In a report published by the NASA Exoplanet Exploration Program (Mamajek & Stapelfeldt), a list of 160 target stars was released as candidates for further study due to various factors such as albedo, habitable zone limits, magnitude, and binarity. All of the stars selected were quite close, and needed to be the right magnitude such that the telescope can differentiate between the star and the exoplanets. We wanted to see what trends and properties we could see in this group of stars, in order to get a better picture of what the Habitable Worlds Observatory was working with and perhaps provide some insight as to what data would be needed so that we would have all the information we need before the HWO launches, such as factors like binarity and magnitude, that affect the ease with which we can study the star systems.

Much previous research has been done regarding characterizing stars. The characterization of stars involves the detailed description of several traits, such as magnitude, composition, and light curves. An example can be seen in the characterization of the two M dwarves K2-18 and TOI-732 (Sairam & Madhusudhan), where the previously described characteristics were described in detail. The metallicities, temperatures, and stellar activities were predicted through using specific spectral lines. The rotation period was obtained through photometric observations. The ages were estimated using the rotation speeds, as rotation slows

down over time for M dwarves. We set out to apply similar methods, finding available photometric, spectroscopic, and temperature data for the stars in the HWO potential target list.

Gyrochronology is a popular method for determining the ages of stars. Gyrointerp is a Python package that can be used to find ages of stars with values of effective temperature and rotation period (Bouma et al.). It has several limitations, such as only being able to predict ages under 4 Gyr and being limited to F, G, and K dwarves. Rotation data and ages of potential HWO targets have been compiled into a catalog called the Activity and Rotation Catalog (ARC), and the data was obtained from “HWO Target Stars and Systems: Activity and Rotation Catalog (ARC) of Potential Target Stars for the Habitable Worlds Observatory”. Characterization has also been done for some of the HWO target stars, with properties such as metallicities, temperatures, masses, and such being analyzed in depth (Harada et al.). However, not much work has been done for analyzing the ages of the specific 160 stars listed in the smaller target list, so we set out to verify these ages using gyrochronology and test whether gyrochronology is effective for these types of stars.

Often, data can't directly be gathered from stars for various reasons. In these cases, using tables with empirical predictions is useful. We planned to use one of these tables, Table 5 in "Intrinsic Colors, Temperatures, and Bolometric Corrections of Pre-main-sequence Stars" (Pecaut & Mamajek), to help fill in gaps in our data. We also looked to compare the available data with the predicted values to see how accurate the empirical predictions were, testing the quality of said predictions.

We predicted that the stars would be typically Sun-like, with a few binaries, as this was what was described in the NASA Exoplanet Exploration Program report (Mamajek & Stapelfeldt). We also predicted that the stars would have ages similar to that of the Sun. Finally,

we predicted that the actual values would be close to the predicted values, though there might be some slight variation.

Astronomers use many digital methods to process and display data. We planned to use tools in Python such as Simbad, Xmatch, and astroquery in order to match the stars to the data already available about them, then use these properties to discover trends. We planned to analyze the similarities and differences between stars by plotting these properties against each other, such as magnitude and temperature. In addition, plotting the predicted values on the same chart would allow us to see how close the predicted values were to the true values.

Method

Sample

Star IDs were taken from the HWO target list (Mamajek & Stapelfeldt), and there are 164 stars for which IDs were provided. There were 47 Tier A stars, 51 Tier B stars, and 66 Tier C stars. The tiers were ordered by how good they were for further study, with Tier A being the best and Tier C being the worst. The class A stars had the right amount of angular separation between the exoplanets and the stars and had the right magnitudes for studying them easily. The B or C tiers all had some traits that made them less suitable, such as binaries or rings.

The table titled "A Modern Mean Dwarf Stellar Color and Effective Temperature Sequence" (Pecaut & Mamajek), an empirical prediction table commonly used for predicting values for stars that have some known values, was also used. The data for the table was gathered from main sequence stars in young moving groups nearby, so it was expected to fit the SIMBAD data quite well as most of the stars in the HWO potential target list are main sequence stars.

The activity and rotation catalog, or ARC, from "HWO Target Stars and Systems: Activity and Rotation Catalog (ARC) of Potential Target Stars for the Habitable Worlds Observatory" was also used to find recorded rotation periods and ages for the stars in the HWO list.

Procedure

The star IDs were uploaded into a table and then uploaded to Python as a csv file. The Simbad program was used to match the star IDs to their positions and also obtain magnitudes, Fe/H ratios, and spectral classes. Using parallax data, the absolute magnitudes were calculated.

The absolute G magnitudes were plotted as box-and-whisker plots and also as histograms to find out the differences between the tiers (see Figures 1 and 2). The Fe/H ratios, obtained from

Simbad, and the ages, and rotation periods, obtained from the ARC, were also plotted as box-and-whisker plots. A color magnitude diagram was also created by plotting the B-V magnitude against the G magnitude.

The prediction table was also uploaded as a csv file. The positions obtained from Simbad were cross matched with the Gaia dataset with Xmatch, and thus temperatures and other magnitudes were obtained. The matching radius was determined using graphs of matches vs. angular distance, and checking the cutoff point where matches seemed to fall dramatically. In this manner, an exploratory matching radius of 5 arcseconds was decided on. Various graphs were then plotted with the prediction table against the temperature and magnitude, notably temperature magnitude diagrams, color magnitude diagrams, and color-color diagrams.

The star coordinates were used to find the stars' TIC IDs. These IDs were crossmatched with the ARC and the TESS input catalog, and a table was created with the stars and the rotation periods and ages. Then, the gyrointerp program was used to find the ages for the stars that had recorded ages younger than 4 Gyr through gyrochronology, and these calculated ages were compared with the recorded ages.

Results

The stars tended to be quite Sun-like in magnitude, with magnitudes around the Sun's value of 4.83. Although they did tend to be Sun-like, in the graphs, there was a bit of variation, especially among the A set, which tended to be dimmer. The B and C sets tended to be brighter. A histogram of the magnitudes is presented in Figure 1 and a box-and-whisker plot is presented in Figure 2.

As for spectroscopy, the stars also tended to be Sun-like, with Fe/H ratios of close to 0. Once again, Tier A stars seemed to be more variable. However, Tier C also had a lot of variability, with many old outliers. Tier A stars generally had higher Fe/H ratios, while Tier B and C stars tended to have lower ones. A box-and-whisker plot is presented in Figure 3.

Color-magnitude, color-temperature, and color-color diagrams were also plotted, provided in Figure 4, 5, 6, and 7. The data tended to follow the prediction table mentioned earlier, but there were some slight discrepancies.

The ages of the stars had a median of around 5.5 Gyrs, older than that of the Sun's, with the A and B stars being older while the C stars were younger. This was quite interesting because it seemed to go against the trends in metallicity. A box-and-whisker plot is presented in Figure 8. The ages for the A stars were more variable than those of the B and C stars. As for rotation period, the A and B stars tended to be quite similar, with C stars rotating slightly faster. The A stars were once again the most variable. A box-and-whisker plot is presented in Figure 9.

Finally, the results with gyrochronology varied greatly. There seemed to be no pattern between the predicted age and the recorded age. A scatterplot of these values is presented in Figure 10.

Discussion

We were able to find out that most stars were Sun-like, with a few discrepancies, such as giants and binaries, which were expected given the selection process of the stars. They all tended to be quite bright and well-studied, as they were picked as good candidates for exoplanet detection. However, the ages of the stars tended to be a bit older than that of the Sun. Thus, the stars mostly lined up with our expectations.

The Tier A group tended to have stars with more differing properties, but they tended to be more Sun-like than the other groups, with the exception of age. The Tier B and C groups had a lot of outliers in terms of Fe/H ratios, but they tended to be less widespread. However, they tended to be systematically brighter, thus suggesting the presence of many binaries. This was also in line with the expectation in terms of binarity and suitability for further study; the A stars tended to be more suitable as targets as they were more Sun-like.

An interesting finding was the discrepancy between the prediction table predicted values and the actual observed values. In the color-magnitude diagram, the prediction table consistently predicted dimmer values, likely due to the presence of binaries. In the color-temperature diagram, the prediction table also underpredicted temperature. This was interesting, because this discrepancy was consistent and for stars of all magnitudes. It could perhaps be explained by measurement technique, but a constant overprediction of 50 K is quite interesting and something to be investigated further.

The Simbad data had good coverage for the stars in the catalog, but using Xmatch with the Gaia catalog didn't yield that many results. This was a major issue in our research, as the Gaia data was needed to progress further. Thus, the Habitable Worlds Observatory could focus

on gathering basic data for all the stars first, such as the different magnitudes and temperatures. This would build a good foundation for further research regarding the stars in the dataset.

Also, the gyrointerp program didn't work well for stars in the ARC, thus discouraging us from applying it to the stars in the HWO catalog. This could be due to multiple reasons; first, the rotation periods published in the ARC might have inaccuracies, which would lead to differing calculated ages. In addition, some of the stars in the ARC may be binaries or in a spectral class that is too hot for these calculations, although we did our best to filter these out. Even considering these factors, this discrepancy is extremely large and deserves further investigation with different datasets.

Another future research direction could be using photometry for gathering light curves for the stars or using more spectroscopy to find out compositions. These paths would also yield valuable data for exoplanet detection and calculations for the likelihood of habitable exoplanets.

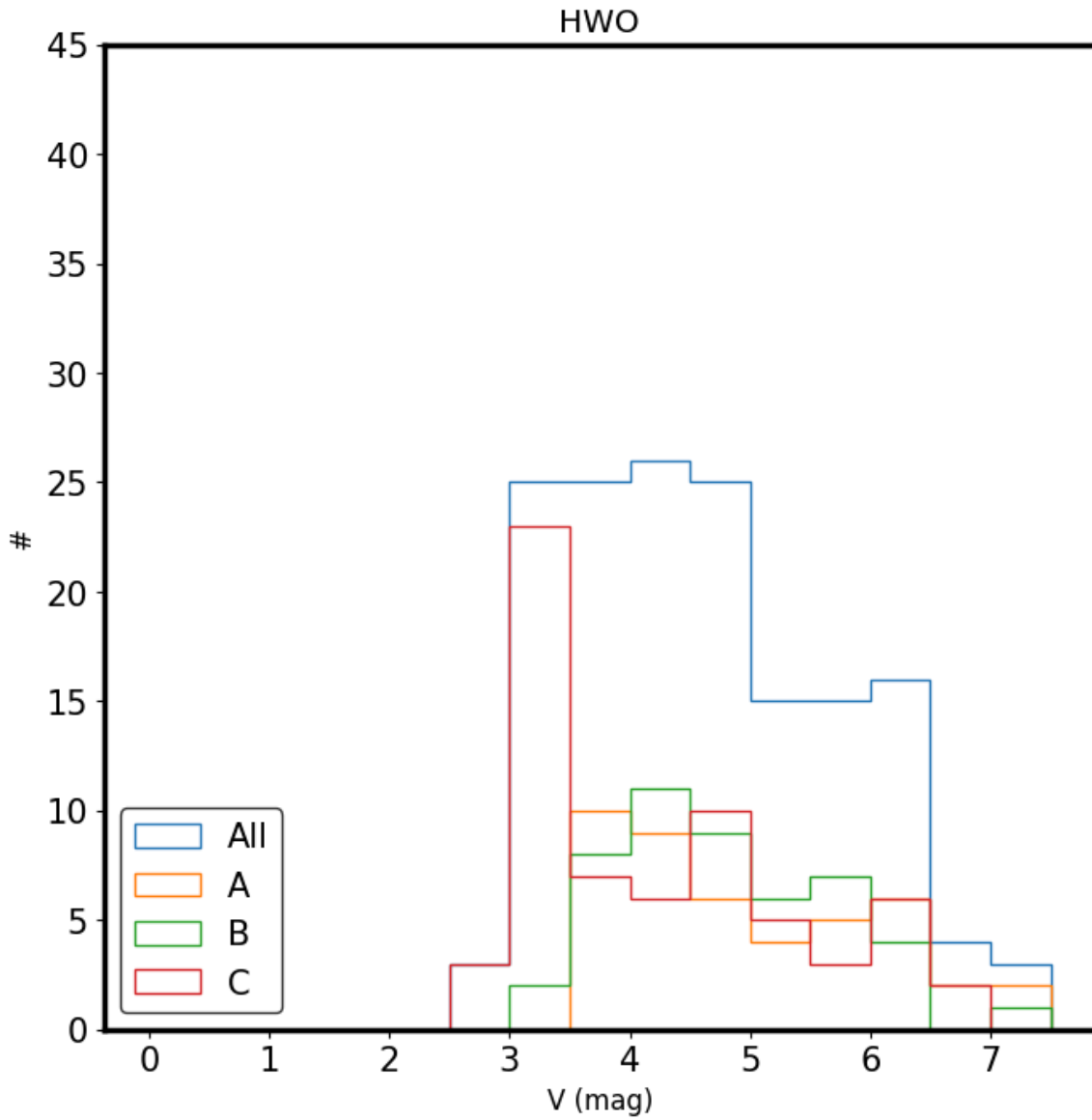
Conclusion

Searching for exoplanets requires vast amounts of data. However, data collection is laborious and often inefficient. Thus, knowing what data to gather is of great importance. Through this study, we examined the data already openly available about the target list of stars and hope that this information can be useful for future characterization research. In addition, gyrochronology is an important tool in astronomy for estimating the ages of stars. However, it seems to be quite limited in stars in the ARC due to various reasons discussed above, which deserves more investigation.

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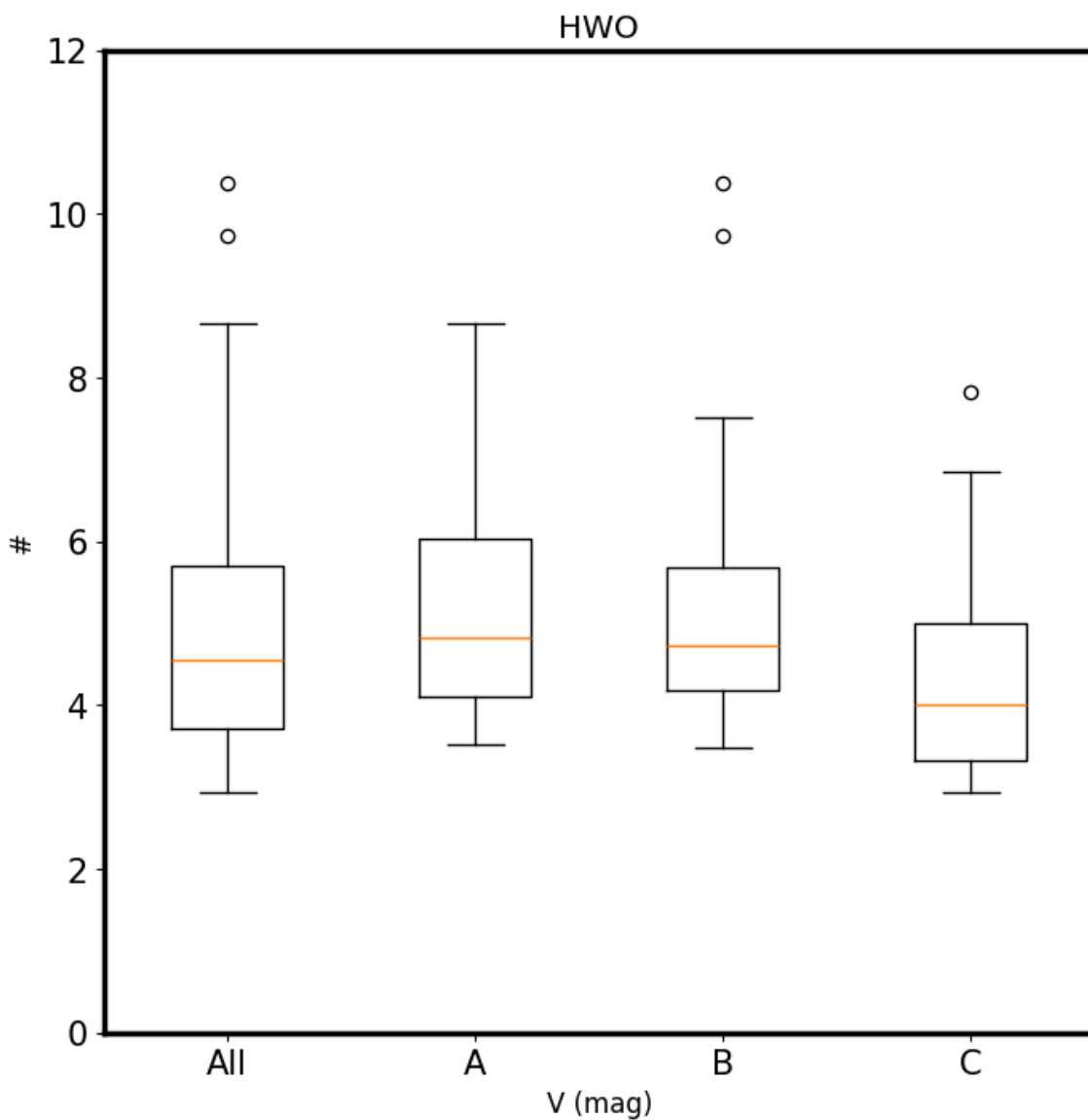
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Figure 1*Histogram of HWO absolute magnitudes*

Note: Tier A and Tier B stars tended to be more spread out, but dimmer and closer to Sun-like values. Tier C stars tended to be brighter, perhaps due to binaries.

Figure 2

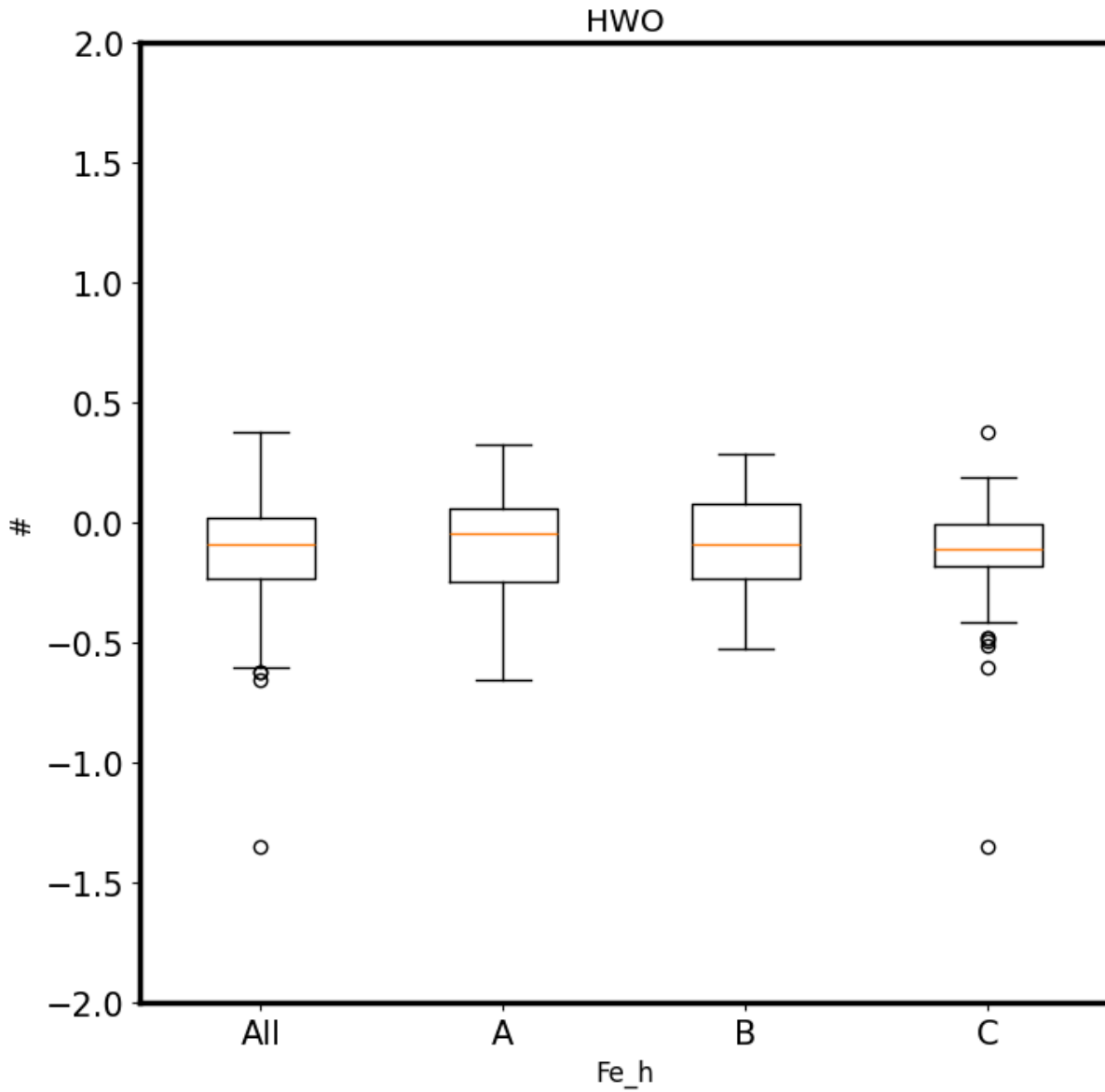
Box-and-whisker plot of HWO absolute magnitudes



Note: The same results as Figure 1 can be seen here.

Figure 3

Box-and-whisker plot of Fe/H ratios

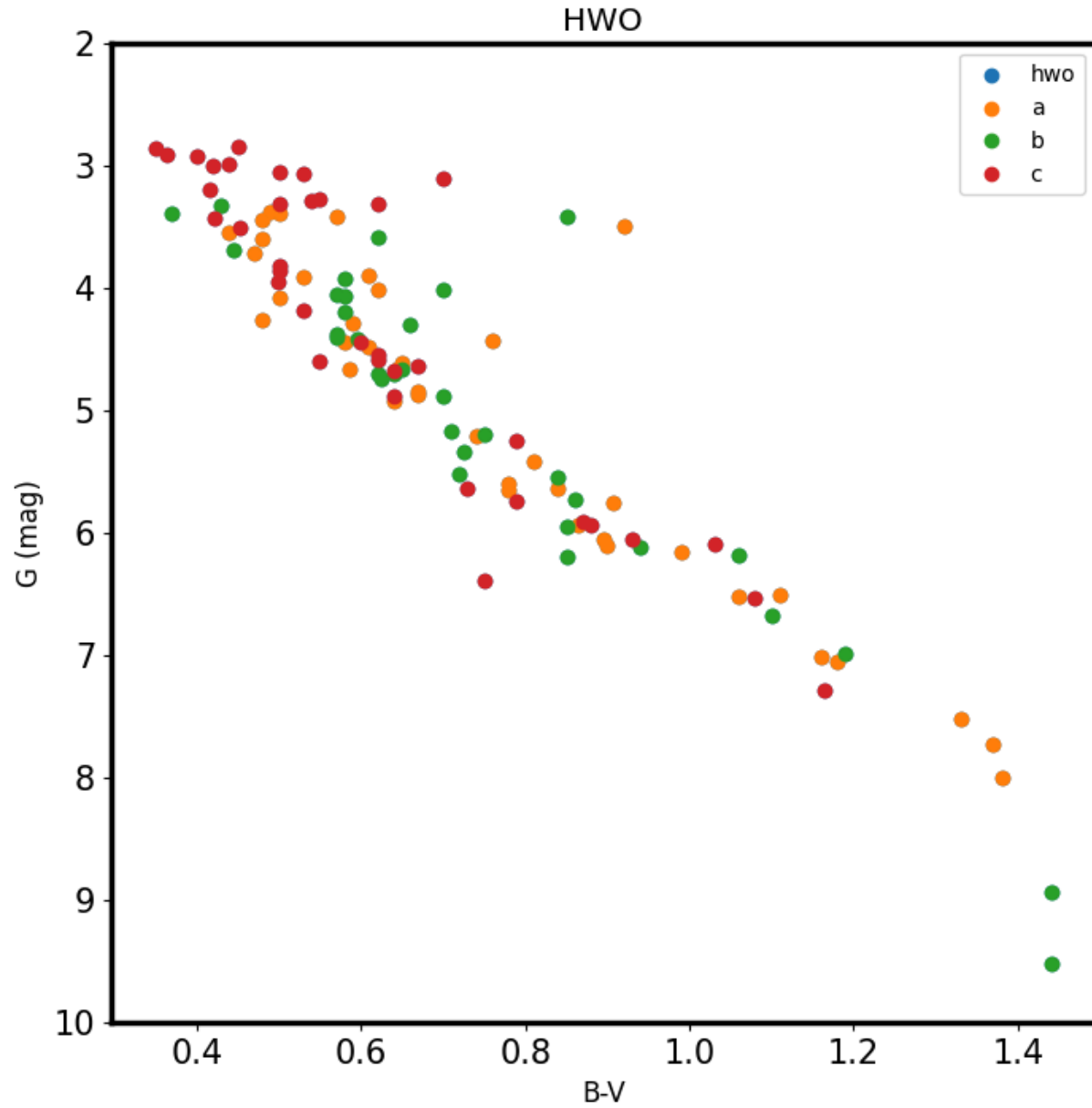


Note: The Sun has a value of 0. Tier A has higher values, while B and C have lower values.

There are a lot of outliers for Tier C stars.

Figure 4

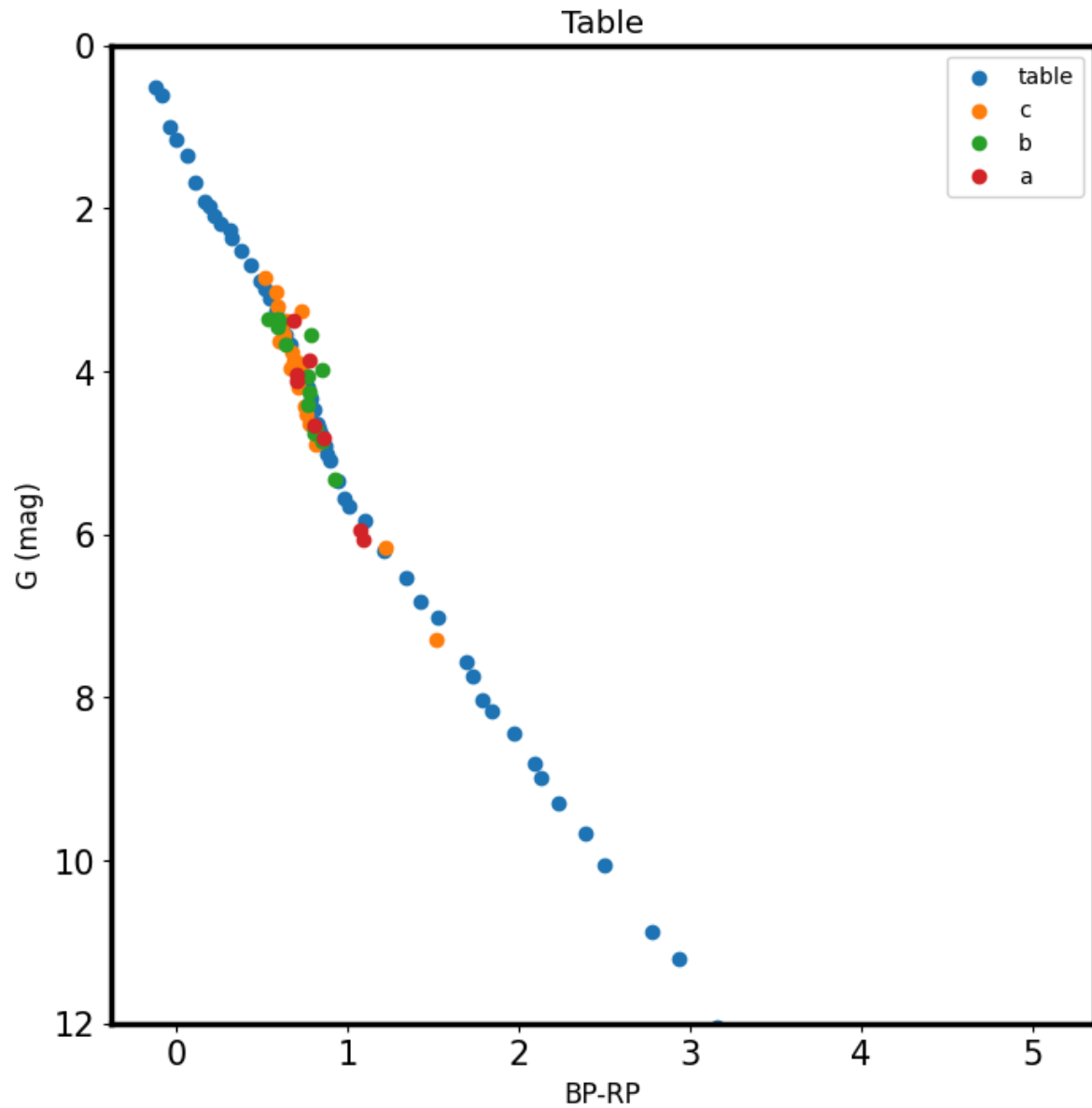
Color-magnitude diagram with G magnitudes plotted against $B-V$ magnitudes using Simbad data



Note: A strong main sequence can be seen, but there seems to also be some red giants and binaries due to the presence of the widening in the upper left portion of the graph. The labels a, b, and c represent the Tier A, Tier B, and Tier C stars, respectively.

Figure 5

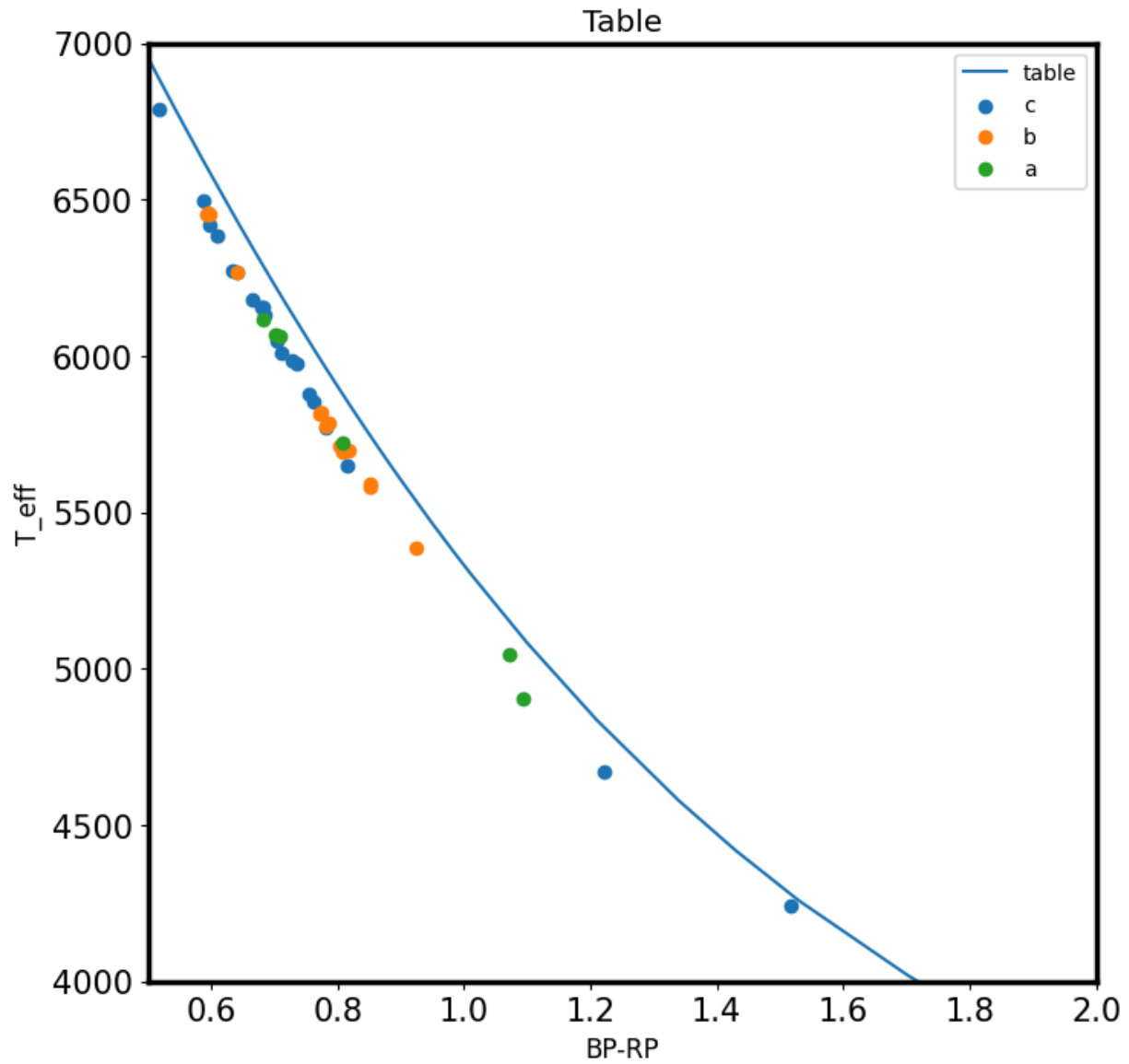
Color-magnitude diagram with G magnitudes plotted against $BP-RP$ magnitudes using Gaia data and prediction table



Note: The trends in Figure 4 can be seen in this table as well. The prediction table tended to underpredict brightness, likely because of binaries. The table label represents the prediction table data.

Figure 6

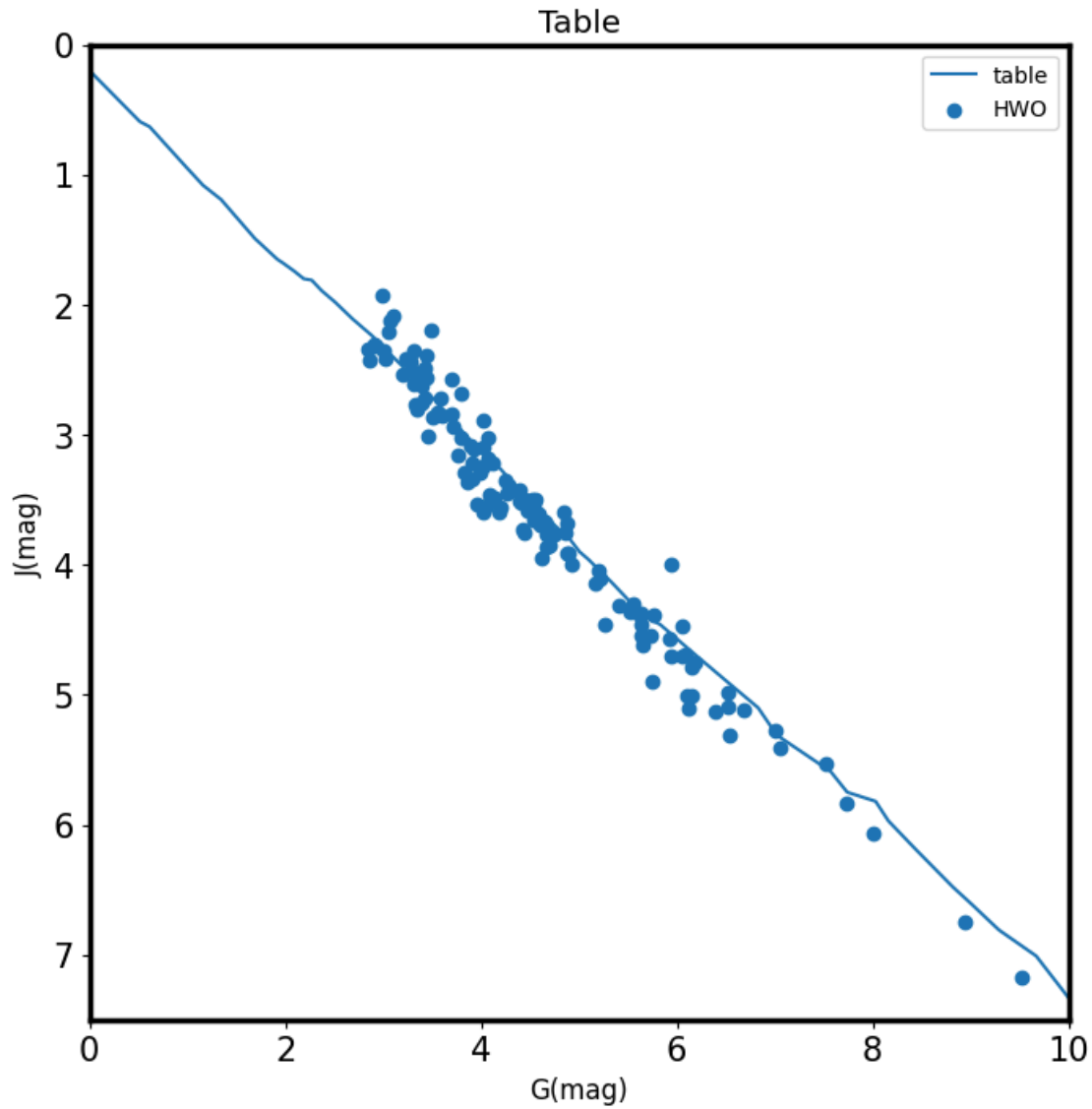
Color-temperature diagram with temperature plotted against BP-RP magnitudes from Gaia data and prediction table



Note: The temperatures closely followed the predicted values from the BP-RP color, but there was a constant discrepancy of around 50 K between the two values, likely due to a difference in measurement methods. The table label represents the prediction table data.

Figure 7

Color-color diagram with J magnitude plotted against G magnitude using data from Simbad and prediction table



Note: The prediction table tended to slightly overpredict J magnitudes, but the real data closely followed the expected data. The table label represents the prediction table data, while the HWO label represents the data from Simbad.

Figure 8

Box-and-whisker plot of recorded ages (Gyr)

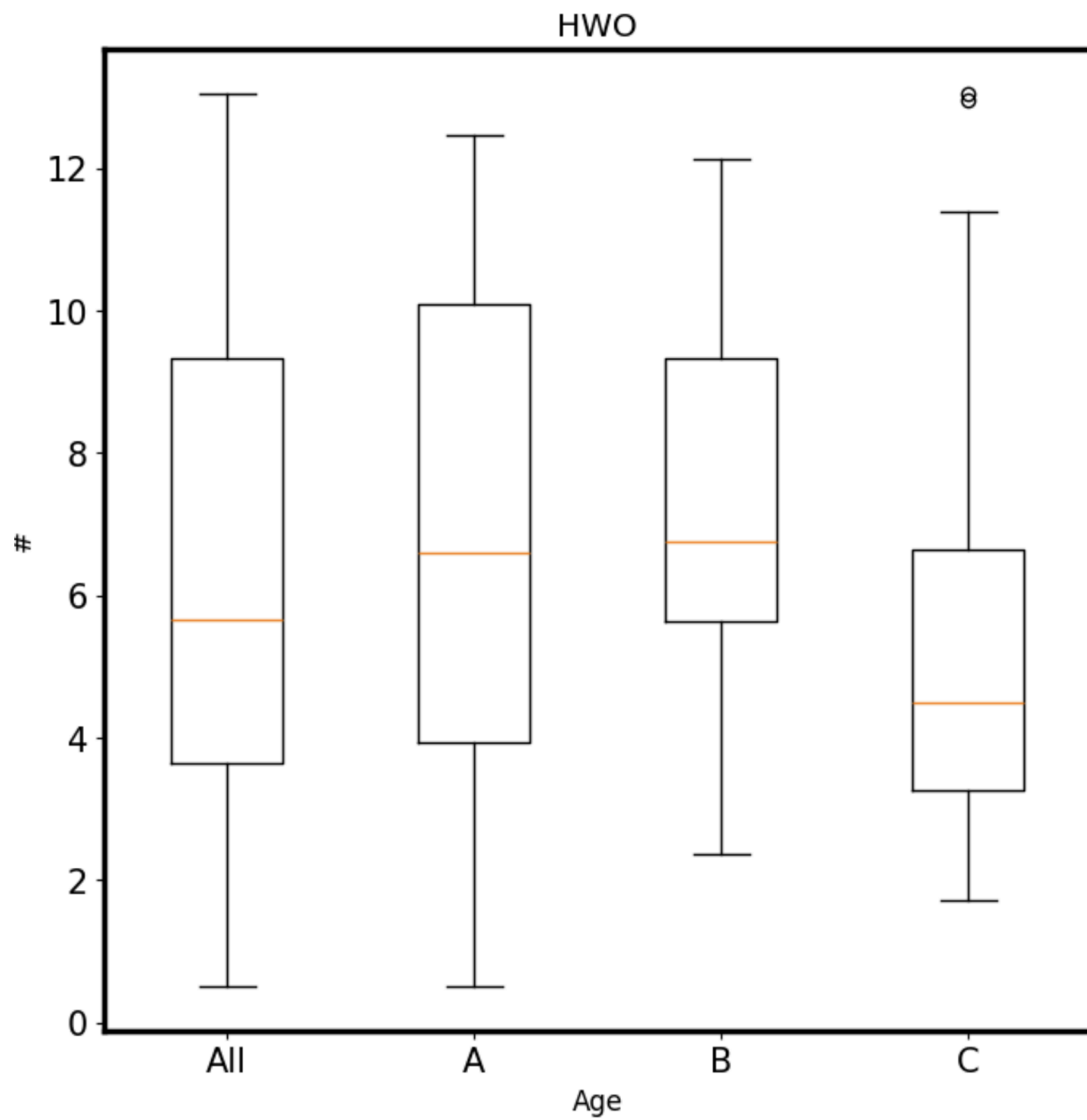


Figure 9

Box-and-whisker plot of recorded rotation periods (days)

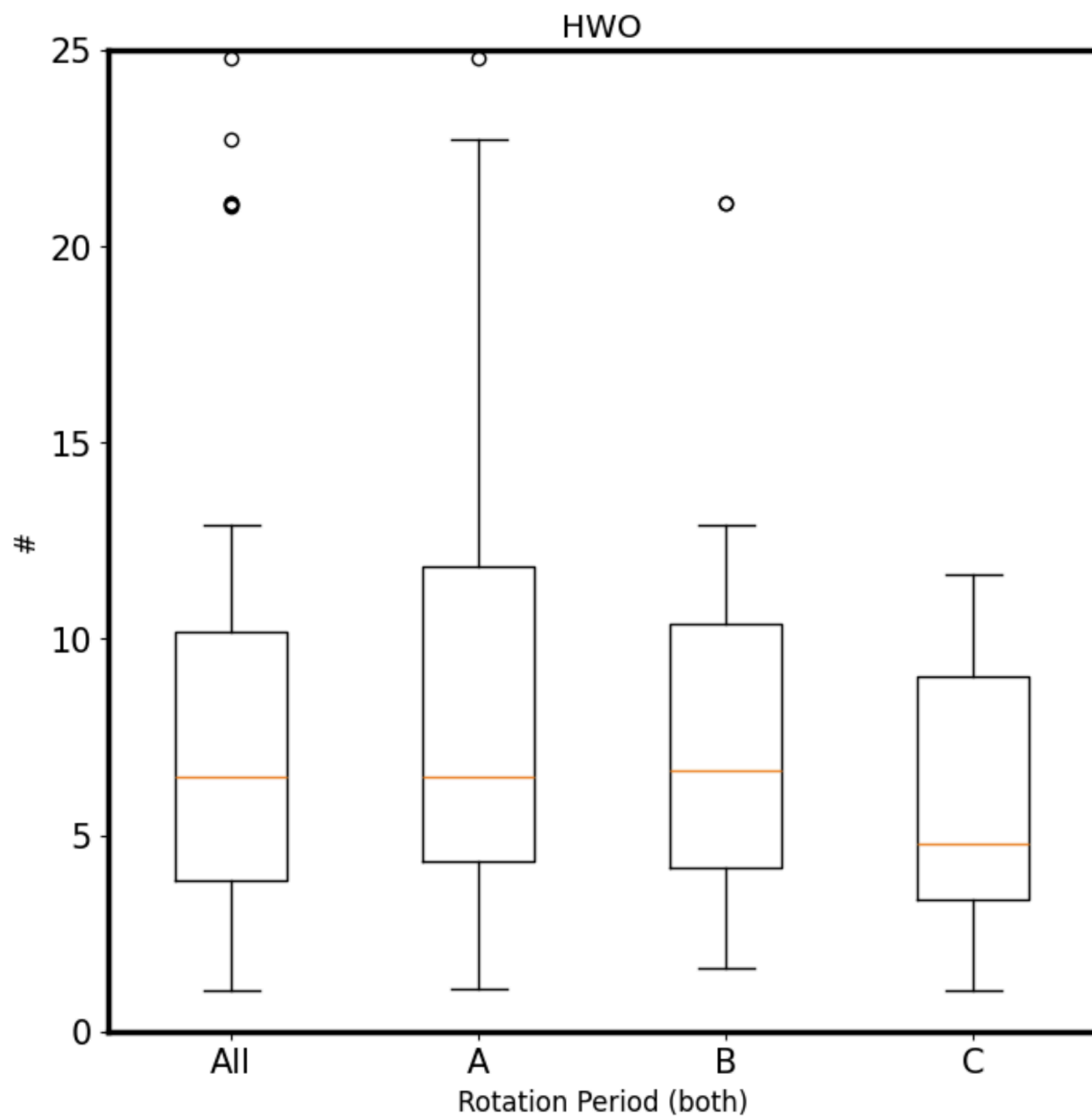


Figure 10

Scatterplot of recorded age (Gyr) vs. calculated age (Gyr)

